Dr Maria Zamyatina — CV

Contact

m.zamyatina@exeter.ac.uk | mzamyatina | mzamyatina.com

Academic career

Postdoctoral Research Fellow — Aug 2025-now
Department of Physics and Astronomy, University of Exeter | Exeter, UK
Maternity leave (9 months)

Postdoctoral Research Fellow — Apr 2022-Jul 2025
Department of Physics and Astronomy, University of Exeter | Exeter, UK

Postdoctoral Research Fellow — Sep 2019-Mar 2022
Department of Physics and Astronomy, University of Exeter | Exeter, UK

Education

PhD in Environmental Sciences — 2015-2020
School of Environmental Sciences, University of East Anglia | Norwich, UK
Supervisors: Prof. Claire Reeves, Dr Paul Griffiths, Dr Marcus Köhler, Dr Mike Newland
Thesis: Impacts of C_1-C_3 alkyl nitrates on tropospheric ozone chemistry

MSc in Climate Change with Distinction — 2014-2015
School of Environmental Sciences, University of East Anglia | Norwich, UK
Supervisor: Prof. Claire Reeves
Thesis: Investigation of the relationship between tropospheric ozone production efficiency and carbon bond emissions

Specialist Diploma in Meteorology — 2009-2014
Faculty of Geography, Lomonosov Moscow State University | Moscow, Russia
Supervisor: Prof. Alexander V. Kislov
Thesis: Climatically-induced variations of the Caspian Sea level over the last Millennium

Publications

19. Steinrueck M. E., Savel A. B., Christie D. A., Carone L. et al. (incl. **Zamyatina M.**) (submitted). Limb Asymmetries on WASP-39b: A Multi-GCM Comparison of Chemistry, Clouds, and Hazes.
18. Mukherjee S., Sing D. K., Fu G., Stevenson K. B. et al. (incl. **Zamyatina M.**) (submitted). Cloudy mornings and clear evenings on a giant extrasolar world.
17. Ahrer E-M., Fairman C., Kirk J., Wakeford H. R. et al. (incl. **Zamyatina, M.**) (2025). BOWIE-ALIGN: Weak spectral features in KELT-7b's JWST NIRSpec/G395H transmission spectrum imply a high cloud deck or a low-metallicity atmosphere. MNRAS.
16. Fu G., Mukherjee S., Stevenson K. B., Sing D. K. et al. (incl. **Zamyatina M.**) (2025). Overcast mornings and clear evenings in hot Jupiter exoplanet atmospheres. ApJL.
15. Mak M. T., Sergeev D. E., Mayne N. J., **Zamyatina M.**, Steinrueck M. E. et al. (2025). The impact of different haze types on the atmosphere and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b and WASP-39b. MNRAS.
14. Ahrer, E-M., Gandhi, S., Alderson, L., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). Tracing the formation and migration history: molecular signatures in the atmosphere of misaligned hot Jupiter WASP-94Ab using JWST NIRSpec/G395H. MNRAS.
13. Meech, A., Claringbold, A. B., Ahrer, E-M., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). BOWIE-ALIGN: Sub-stellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy. MNRAS.
12. Kirk, J., Ahrer, E-M., Claringbold, A. B., **Zamyatina, M.**, Fisher C. et al. (2025). BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b. MNRAS.
11. Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions. MNRAS.
10. Kirk, J., Ahrer, E-M., Penzlin, A. B. T., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history. RAS Techniques and Instruments.
9. Espinoza, N., Steinrueck, M., Kirk, J., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2024). Inhomogeneous terminators on the exoplanet WASP-39 b. Nature.
8. Christie, D. A., Mayne, N. J., **Zamyatina, M.**, Baskett, H., Evans-Soma, T. M., et al. (2024). Longitudinal filtering, sponge layers, and equatorial jet formation in a general circulation model of gaseous exoplanets. MNRAS.
7. **Zamyatina, M.**, Christie, D. A., Hébrard, E., Mayne, N. J., Radica, M. et al. (2024).

Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example. MNRAS.

6. Taylor, J., Radica, M., Welbanks, L., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2023). Awesome SOSS: atmospheric characterisation of WASP-96b using the JWST early release observations. MNRAS.
5. Radica, M., Welbanks, L., Espinoza, N., Taylor, J. et al. (incl. **Zamyatina, M.**) (2023). Awesome SOSS: transmission spectroscopy of WASP-96b with NIRISS/SOSS. MNRAS.
4. **Zamyatina, M.**, Hébrard, E., Drummond, B., Mayne, N. J., Manners, J. et al. (2023). Observability of signatures of transport-induced chemistry in clear atmospheres of hot gas giant exoplanets. MNRAS.
3. Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A. et al. (incl. **Zamyatina, M.**) (2022). CAMEMBERT: A Mini-Neptunes GCM Intercomparison, Protocol Version 1.0. A CUISINES Model Intercomparison Project. Planet. Sci. J.
2. Ridgway, R. J., **Zamyatina, M.**, Mayne, N. J., Manners, J., Lambert, F. H. et al. (2023). 3D modelling of the impact of stellar activity on tidally locked terrestrial exoplanets: atmospheric composition and habitability. MNRAS.
1. Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., **Zamyatina, M.** et al. (2022). Lightning-induced chemistry on tidally-locked Earth-like exoplanets. MNRAS.

Invited talks

Jul 2025 Atmospheric chemistry of solar and extrasolar gas giants (review)
Exoclimes VII conference | Montreal, Canada

Mar 2024 Overview of the Met Office Unified Model configuration for hot Jupiter atmospheres
International Space Science Institute (ISSI) workshop | Bern, Switzerland

Feb 2024 Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere
University of Bristol (astronomy seminar) | Bristol, UK

Feb 2023 Atmospheric dynamics and chemistry on exoplanets
University of Queensland (astronomy seminar) | Brisbane, Australia
University of Southern Queensland (exoplanet seminar) | Brisbane, Australia
University of New South Wales (astronomy seminar) | Sydney, Australia

Nov 2022 Observability of signatures of wind-driven chemistry in atmospheres of hot gas giants
Ludwig Maximilian University (exoplanet group seminar) | Munich, Germany
Celebrating JWST's first six months of exoplanet data workshop | Ringberg castle, Germany

Oct 2022 Modelling chemistry of hot Jupiter atmospheres with the Met Office Unified Model
Met Office | Exeter, UK

Feb 2022 Transport-induced quenching shapes transmission spectra of warm and hot Jupiters
University of Warwick (astronomy seminar) | virtual

Contributed talks

Sep 2023 Metallicity masquerade: how to use quenching to distinguish between different planet metallicities
University of Bristol (BOWIE meeting) | Bristol, UK

June 2021 Overview of the Met Office Unified Model adapted to simulate exoplanetary atmospheres
Ariel consortium meeting | virtual

Apr, Sep 2021 3D simulations of warm and hot Jupiter atmospheres: the role of 3D mixing in shaping CH_4-to-CO conversion pathways
EPSC conference | virtual
UKEXOM conference | virtual
University of Exeter (astronomy seminar) | Exeter, UK

Mar, Apr, Jun 2019 Impact of C_1-C_3 alkyl nitrate chemistry on tropospheric ozone: box and global model perspectives
University of Exeter (XCS seminar) | Exeter, UK
EGU conference | Vienna, Austria
University of East Anglia (AMB seminar) | Norwich, UK

Apr 2017 Adding new chemistry into UM-UKCA
Cambridge-EnvEast doctoral alliance symposium | Cambridge, UK

Sep 2012 Assessment of climatological potential of transboundary air pollution transport in Eastern Siberia and the Russian Far East
Air quality management at urban, regional and global scales 4th international symposium/IUAPPA regional conference | Istanbul, Turkey

Posters

Jul 2025 CH_4-CO conversion pathways in hot Jupiter atmospheres
Exoclimes VII conference | Montreal, Canada

Apr, Jun 2024 Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere
UKEXOM conference | Birmingham, UK
Exoplanets 5 conference | Leiden, Netherlands

Sep 2022 Applying known chemical kinetics data to model atmospheres of extrasolar planets

	iCACGP-IGAC conference | Manchester, UK	
Sep 2021	Local and global impacts of C_1-C_3 alkyl nitrate photochemistry and emissions on tropospheric ozone IGAC conference | virtual	
Sep 2018	Impact of alkyl nitrate chemistry on tropospheric ozone iCACGP-IGAC conference | Takamatsu, Japan	
Mar, Apr 2018	Impact of C_1-C_5 alkyl nitrate chemistry on tropospheric ozone - a box modelling study Cambridge-EnvEast doctoral alliance symposium | Cambridge, UK EGU conference | Vienna, Austria	

Awards

2023	Above & Beyond Award	
2022	**EPSRC vacation internship** (for 3 interns)	12893.55£
2022	**Jackson-Grime-Davies (JGD) research internship** (for 1 intern)	2428.71£
2021	**IGAC Early Career Scientist poster prize & travel grant**	1227.70£
2015-2019	**Lord Zuckerman studentship**	112269.50£
2014-2015	**Simon Wharmby postgraduate scholarship**	3000.00£
2012	**World Meteorological Organization travel grant**	1154.10£

Awarded observing time

Feb 2024	JWST's exoplanet grand tour spectroscopic survey	
▪	Co-I HST GO-17612 (PI: David Sing)	24 orbits
▪	Co-I JWST GO-5924 (PI: David Sing)	125.70 hours
Feb 2024	Starspots, hazes, and disequilibrium chemistry: a deep dive into the atmosphere of HAT-P-18b	
▪	Co-I JWST GO-5844 (PI: Michael Radica)	16.40 hours
May 2023	Putting it all together: Dynamics and chemistry probed through transmission spectroscopy of a cloud-free exoplanet	
▪	Co-I JWST GO-4082 (PI: Michael Radica, Co-PI: Jake Taylor)	6.69 hours
May 2023	Hot Jupiter atmospheric forecast: are mornings cloudier than evenings in other worlds?	
▪	Co-I JWST GO-3969 (PI: Nestor Espinoza, Co-PI: Diana Powell)	61.53 hours
May 2023	Does atmospheric composition actually trace formation? Observing aligned vs misaligned hot Jupiters as a testbed	
▪	Co-I JWST GO-3838 (PI: James Kirk Co-PI: Eva-Maria Ahrer)	49.21 hours
May 2023	Testing the C/O ratio prediction for hot Jupiters from disk-free migration	
▪	Co-I JWST GO-3154 (PI: Eva-Maria Ahrer)	10.36 hours

Supervision

Primary supervisor and co-supervisor. Students who went on to do a PhD are marked with *.

PhD supervision (2)

Sep 2024-now	Harry Baskett Thesis: TBD Co-supervisors: Dr. E. Hebrard, Prof. N. J. Mayne
Nov 2020-May 2023	Robert J. Ridgway Thesis: Simulating the impact of stellar flares on the climate and habitability of terrestrial Earth–like exoplanets Co-supervisors: Prof. N. J. Mayne, Prof. F. H. Lambert, Dr. J. Manners

Undergraduate and summer internship supervision (4)

Jun-Aug 2022	EPSRC-funded: Harry Baskett*, Ben Moore*, James McDermott*; JGD-funded: Graig Lils Project: 3D modelling of hot Saturn atmospheric chemistry

Teaching

Jul 2023	**Module leader** Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school | University of Exeter, Exeter, UK
Jul 2022, Jul 2023	**Lecturer** Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school | University of Exeter, Exeter, UK
Sep 2021-Feb 2022	**Associate Tutor** Modules: Experimental science, Frontiers in science University of Exeter | Exeter, UK
Jan 2018	**Instructor** Module: Introduction to Python in Environmental Sciences University of East Anglia | Norwich, UK
2015-2018	**Associate Tutor** Modules: Numerical skills for scientists, Atmospheric chemistry and global change, Atmospheric composition (measurements and modelling), Atmosphere & oceans I

University of East Anglia | Norwich, UK

Academic community

Organisation of scientific meetings

26-30 Jun 2023 Exoclimes VI conference (LOC member, session chair) ~200 attendees
University of Exeter | Exeter, UK

22-24 Jun 2023 ExoSLAM school (LOC member) ~50 attendees
University of Exeter | Exeter, UK

5-6 Dec 2022 BOWIE meeting (co-organiser) 17 attendees
University of Exeter | Exeter, UK

Sep 2017-Jun 2018 Atmospheric and Marine Biogeochemistry (AMB) seminars (co-organiser) ~20 attendees
University of East Anglia | Norwich, UK

Reviewing

Journals: The Astrophysical Journal, Monthly Notices of the Royal Astronomical Society
Proposals: JWST Cycle 3 TAC external expert, JWST Cycle 4 TAC panelist

Outreach

Jul 2024 Joint press release about Espinoza et al. (2024) paper:
Research confirms that distant world's eternal sunrise and sunset are not alike

Sep 2023 Expert scientist at the Climate Exhibition (part of the British Science Festival)

Nov 2015-Jun 2019 Maintainer of @AtmosChemUEA Twitter account

Vocational Experience

Aug-Sep 2013 **Weather Forecaster**
Sheremetyevo International Airport | Moscow, Russia

Jun-Jul 2013 **Technician**
Department of Actinometry, Meteorological Observatory
Lomonosov Moscow State University | Moscow, Russia

Vocational training

Sep 2023 Belbin training
Mar 2023 Leadership training
Dec 2022 Interview training
Sep 2021 Learning and Teaching in Higher Education (LTHE) Unit 1
Mar 2020 JWST proposal planning workshop
Apr 2016 NAME workshop
Jan 2016 Introduction to UKCA
Dec 2015 Introduction to Unified Model
Nov 2015 Introduction to Atmospheric Science
2015-2019 EnvEast Doctoral Training Programme